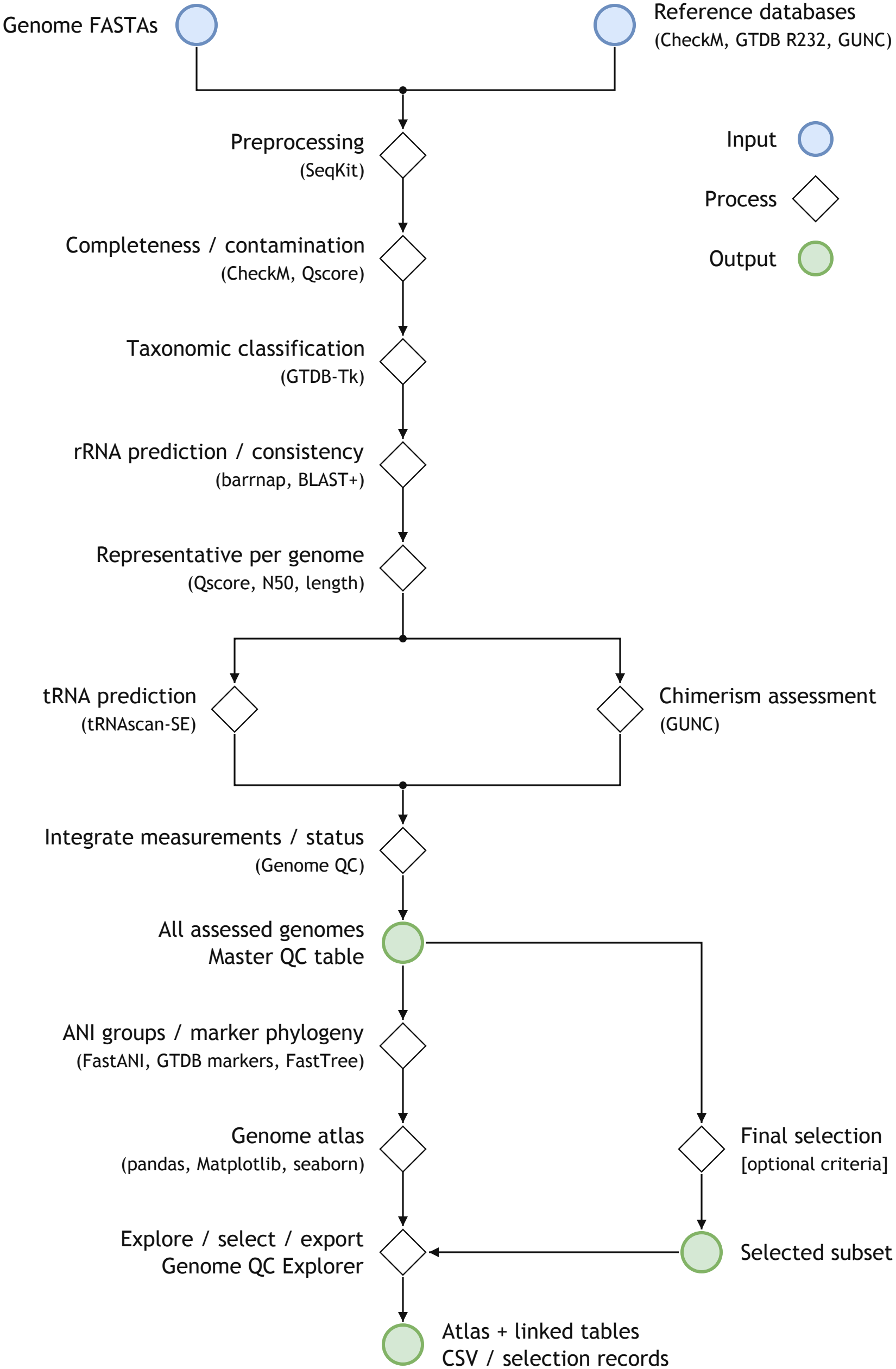
Genome FASTAs
Reference databases
(CheckM, GTDB R232, GUNC)
Input
Process
Output
Preprocessing
(SeqKit)
Completeness / contamination
(CheckM, Qscore)
Taxonomic classification
(GTDB-Tk)
rRNA prediction / consistency
(barrnap, BLAST+)
Representative per genome
(Qscore, N50, length)
tRNA prediction
(tRNAscan-SE)
Chimerism assessment
(GUNC)
Integrate measurements / status
(Genome QC)
All assessed genomes
Master QC table
ANI groups / marker phylogeny
(FastANI, GTDB markers, FastTree)
Genome atlas
(pandas, Matplotlib, seaborn)
Final selection
[optional criteria]
Explore / select / export
Genome QC Explorer
Selected subset
Atlas + linked tables
CSV / selection records